CS 185/285 Assignment 5: Offline RL

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1 SAC+BC

Figure 1: Training Curves of SAC+BC on `antsoccer-arena` and `cube-single`

Results are shown in Fig 1. Best success rates are the runs used with $\alpha = 10$ in `antsoccer-arena`, which reached **32%** success rate, with $\alpha = 300$ in `cube-single`, which reached **100%** success rate multiple times.

2 IQL

Results are shown in Fig 2. In `antsoccer-arena`, all the different α runs reached the same **16%** success rate. In `cube-single`, best run was with $\alpha = 1$, which reached **100%** success rate.

In the experiments, IQL was less sensitive to the choice of alpha and produced more stable performance across different values, while SAC+BC was more sensitive to alpha but achieved a higher best performance when alpha was well tuned. This suggests that IQL is easier to tune in practice, whereas SAC+BC can be more performant but requires more careful hyperparameter selection.

3 FQL

Results are shown in Fig 3. Best success rates are the runs used with $\alpha = 10$ in `antsoccer-arena`, which reached **60%** success rate, with $\alpha = 300$ in `cube-single`, which reached **100%** success rate multiple times.

Figure 2: Training Curves of IQL on `antsoccer-arena` and `cube-single`

Figure 3: Training Curves of FQL on `antsoccer-arena` and `cube-single`